

Security System Project Report

1. Introduction

This project implements a secure authentication and communication system using modern cryptographic techniques.

It ensures user authentication, data integrity, and message security using multiple security layers.

2. System Features

The system includes the following components:

1. User Registration

- Users register with username and password.
- A unique random salt is generated for each user.
- Passwords are hashed using SHA-256 with salt before storing.

2. User Login

- Users login using username and password.
- The system retrieves the stored salt and password hash.
- It verifies the password by re-hashing and comparing.

3. JWT Authentication

The system uses JSON Web Token (JWT) for session management.

- Token contains:
 - Username
 - Role (admin/user)
 - Expiration time
- Token is signed using a secret key.
- Access is granted only with a valid token.

JSON Web Token

4. Token Verification

- The system verifies token validity.
- Handles:
 - Expired tokens
 - Invalid or tampered tokens

5. Message Integrity (HMAC)

- Users send messages to the system.
- The system generates an HMAC using a secret key.
- The receiver verifies message integrity.

HMAC

- If message is modified → system detects tampering.

6. Digital Signature

- RSA public/private key pair is used.
- Messages are signed using the private key.
- Signature is verified using the public key.

Digital Signature

- Ensures authenticity and integrity of messages.

7. Role-Based Access Control

- Admin users have full access.
- Normal users have limited access.
- Access is controlled based on role inside JWT.

3. Security Testing

The system was tested against different attacks:

Token Tampering

- Modified tokens are rejected.

Message Tampering

- Altered messages fail HMAC verification.

Signature Tampering

- Modified signatures fail verification.

4. Technologies Used

- Python
- Flask Framework
- SHA-256 hashing
- JWT authentication
- HMAC
- RSA encryption (cryptography library)

Flask

5. Conclusion

This project demonstrates a complete secure authentication system that integrates multiple cryptographic techniques to ensure:

- Secure user authentication
- Data integrity
- Message authenticity
- Role-based authorization

The system successfully detects any unauthorized modifications or tampering attempts.